



APEX AI
SOLUTIONS

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Powered by Innovation*

Mission Statement

To build intelligent, scalable, and accessible AI-driven
platforms that empower businesses

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Use-cases for AI in Hospitality



Fraud & Anomaly Detection in Bookings & Payments



Brief of the Idea & Problem

Detect suspicious or fraudulent bookings and transactions in real-time using machine learning and rule-based automation to reduce chargebacks, no-shows, and fake bookings.

Example Scenario:

Issue Faced:

- Surge in no-show bookings from the same IP using different emails.
- Last-minute cancellations using high-risk credit cards.
- Unexpected spike in multi-room bookings during off-season.



Approach and Impact

1. **ML Models** (e.g., Isolation Forest, XGBoost) analyze booking patterns (IP, time, card BIN, email domain) to assign fraud risk scores.
2. Rules Engine flags known patterns (e.g., disposable emails, VPN usage, bulk bookings).
3. Automation Tools (e.g., UiPath) auto-cancel high-risk bookings, trigger guest verification, and alert staff.
4. Dashboards (Power BI) track trends and model performance for continuous improvement.

Example:

A 3 AM booking for 4 rooms using a mailinator email and VPN IP scores 0.91 risk. System auto-cancels it, saving ~\$2,000.

Impact:

⬇️ 60% fraudulent bookings • ⬇️ 40% chargebacks • ⬆️ Operational efficiency



Predictive Maintenance in Hospitality



Brief of the Idea & Problem

- Hotels rely on critical equipment like HVAC, elevators, boilers, and laundry machines.
- Unexpected failures disrupt guest experience and incur high emergency repair costs.
- Traditional maintenance is either reactive (too late) or scheduled (often unnecessary).
- Predictive maintenance uses AI to forecast equipment failure before it happens, enabling just-in-time servicing, reducing downtime, and extending asset life.



Approach

- **Data Collection:** IoT sensors monitor temperature, vibration, energy usage, etc.
- **Analytics & ML Models:** Algorithms (e.g., time series models, anomaly detection, LSTM) predict failure patterns.
- **Automated Alerts:** Maintenance teams receive early warnings via dashboards or mobile apps.
- **Optimization Loop:** Models improve with more data and technician feedback.



Example Scenario

A hotel's central HVAC system shows abnormal vibration and energy spikes.

The ML model predicts a potential fan motor failure in 5 days.

- An automated alert is triggered via the building management system (BMS).
- A technician inspects and replaces a worn bearing before failure, avoiding a total outage and saving \$3,000 in emergency repair and guest compensation.



RPA + ML for Anomaly Handling in Hotel Operations



Brief of the Idea & Problem

- Hotel operations involve many repetitive, data-heavy tasks — from guest check-in to room assignment and housekeeping coordination.
- Manual anomaly detection (e.g., overbooked rooms, missed cleanings, billing errors) is slow, error-prone, and reactive.
- Combining Machine Learning with Robotic Process Automation (RPA) allows hotels to proactively detect anomalies and automatically trigger responses, reducing disruption and improving guest experience.



Approach

ML for Anomaly Detection

- Detect unusual operational patterns (e.g., sudden delays, data mismatches)
- **Tools:** *Scikit-learn, TensorFlow, PyCaret, Azure ML*

RPA for Response Automation

- Bots handle alerts, escalate issues, or take corrective action
- **Tools:** *UiPath*

Integration

- Connects with PMS, housekeeping systems, and billing platforms via APIs
- **Tech:** *REST APIs, Zapier, or direct PMS integrations (e.g., Cloudbeds, Opera)*

Monitoring & Feedback Loop

- Dashboards for incident trends and false positives
- **Tools:** *Power BI, Grafana, Looker*



Example Scenario

A guest books a room at 4 PM, but the housekeeping system still shows it as “dirty” at 5 PM.

- ML detects this timing mismatch as an operational anomaly.
- RPA bot cross-checks with the room staff’s app and finds the status wasn't updated.
- The bot notifies the supervisor, reassigns cleaning if needed, and sends a status update to the front desk — before the guest arrives.

→ **Result:** No delay at check-in, reduced front desk load, and improved guest satisfaction.



Guest Sentiment & Review Analysis



Brief of the Idea & Problem

- Hotels receive thousands of reviews across platforms (Google, TripAdvisor, OTA portals).
- Manually analyzing this feedback is slow, inconsistent, and reactive.
- AI-based sentiment analysis helps automatically interpret guest opinions, uncover hidden issues, and guide service improvements all in real time and at scale.



Approach

Text Preprocessing & Sentiment Analysis

- Extract sentiment, emotion, and topic from guest reviews and survey data
- **Tools:** *spaCy, NLTK, TextBlob, Hugging Face Transformers*

Topic Modeling & Keyword Clustering

- Identify frequent concerns (e.g., “slow check-in”, “clean rooms”)
- **Tools:** *BERTopic, LDA, GPT embeddings*

Dashboards & Alerts

- Visualize trends and trigger alerts for negative spikes
- **Tools:** *Power BI, Tableau, Looker*

Integration with CRM & PMS

- Feed insights into loyalty campaigns, staff feedback, and escalation workflows
- **Tech:** *REST APIs, Zapier, CRM connectors (Salesforce, Zoho)*



Example Scenario

Over a weekend, a hotel receives 20+ reviews across OTA platforms mentioning "rude front desk" and "long wait times".

1. Sentiment model detects a sharp drop in service tone.
2. Topic analysis clusters these under “check-in experience”.
3. An alert is sent to the operations manager.

→ **Response:** Staff shift is adjusted, and a quick service training session is deployed before it impacts the weekly rating average.



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Thank You